S100运行D457（首次）

一、环境及驱动安装

1. 系统要求：4.0.4-Beta
2. D457*2驱动安装：

linux-image-rdk-s100_6.1.112-rt43-DR-4.0.4-2511271114-⋯
20.66MB

linux-headers-6.1.112-rt43_6.1.112-rt43-DR-4.0.4-⋯
13.21MB

hobot-camera_4.0.4-20251125165934_arm64.deb
12.59MB

下载上述的包放在S100的/userdata目录，以下命令安装

代码块

```
dpkg -i linux-headers-6.1.112-rt43_6.1.112-rt43-DR-4.0.4-2511251651-g43dde6-g5156eb-148_arm64.deb
dpkg -i linux-image-rdk-s100_6.1.112-rt43-DR-4.0.4-2512022103-g147f31-gfc0585-2_arm64.deb
dpkg -i hobot-camera_4.0.4-20251125165934_arm64.deb

```

3. D457*4驱动安装：

linux-image-rdk-s100_6.1.112-rt43-DR-4.0.4-2512181120-⋯
20.01MB

hobot-camera_4.0.4-20251210151549_arm64.deb
12.63MB

下载上述的包放在S100的/userdata目录，以下命令安装

代码块

```
dpkg -i hobot-camera_4.0.4-20251210151549_arm64.deb
dpkg -i linux-image-rdk-s100_6.1.112-rt43-DR-4.0.4-2512181120-g122b1f-dirty-geba25f-dirty-2_arm64.deb
```

4. tros及realsense包安装

代码块

```
//安装或者更新TROS
apt update
apt upgrade
apt install tros-humble

//安装realsense
apt install ros-humble-librealsense2
apt install ros-humble-realsense2-camera
```

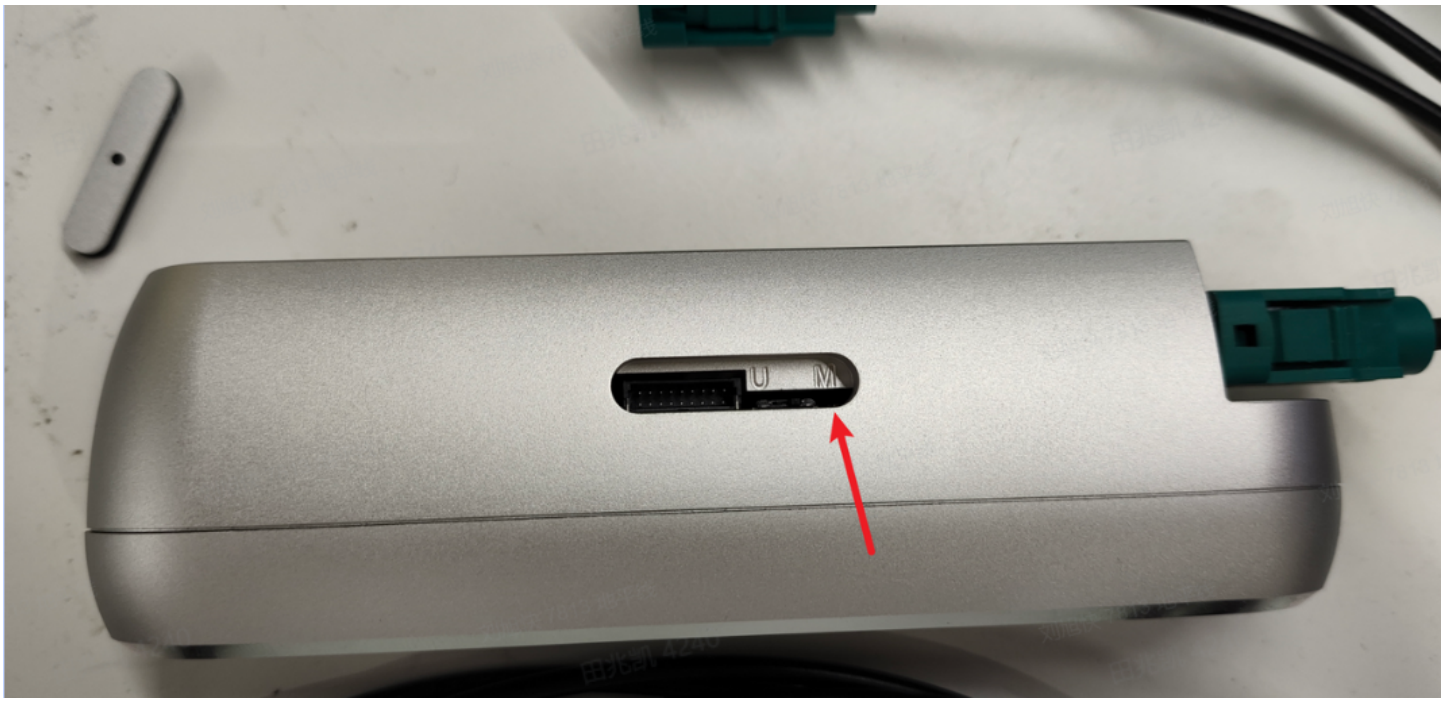
5. 更新librealsense.so

librealsense2.so.2.56.5
12.31MB

代码块

```
//拷贝librealsense2.so.2.56.5到/opt/ros/humble/lib/aarch64-linux-gnu目录
//建立软连接
rm librealsense2.so.2.56
ln -s librealsense2.so.2.56.5 librealsense2.so.2.56
```

6. 将相机固件升级至最新版本，需要将拨码播至U状态，通过USB连接至终端设备通过REALSENSE-VIEWER进行升级
7. Realsense D457切换到usb模式，拨码到M，如下图



8. 修改config.txt 文件，配置要添加的dtbo文件
   若该位置没有config.txt文件， 用户可自行添加config.txt 文件

```
root@ubuntu:/boot# ls
System.map-6.1.112-rt43-DR-4.0.3-2508281331-gb851f5-g374a40-dirty  hobot
config                                                              initrd.img-6.1.112-rt43-DR-4.0.3-2508281331-gb851f5-g374a40-dirty
config-6.1.112-rt43-DR-4.0.3-2508281331-gb851f5-g374a40-dirty       lost+found
config.txt                                                          overlays
extlinux                                                            vmlinuz-6.1.112-rt43-DR-4.0.3-2508281331-gb851f5-g374a40-dirty
```

按照下面的方法，修改config.txt

`dtbo_file_path=/overlays/v0p5_d457_2v_depth_color.dtbo` D457*2

`dtbo_file_path=/overlays/d457_4V_RX4_portAC_RX1_portAC_depth_color.dtbo` D457*4

9. 修改 blacklist

   避免 d457 ko 自动加载，板端在 /etc/modprobe.d/blacklist-camsys.conf 文件中，增加

`blacklist d457`

10. 重启系统

二、D457相机*2启动

当前s100中无其他相机设备，若连接了其他相机设备需修改对应设备号

1. 切换到v4l2模式

代码块

```
#卸载hbn驱动
rmmod hobot_isp
rmmod hobot_cim
rmmod hobot_mipidbg
```

```
rmmod hobot_mipicsi
rmmod hobot_pym_jplus
rmmod hobot_gdc
rmmod hobot_ynr

#加载v4l2驱动
modprobe videobuf2-common

modprobe videobuf2-v4l2

modprobe videobuf2-memops

modprobe videobuf2-dma-contig

modprobe v4l2-mem2mem

modprobe max9295a
modprobe max96712
modprobe d457

modprobe v4l_mipicsi

modprobe v4l2_cim

modprobe hobot_isp_v4l2

modprobe pym_v4l_drv
modprobe gdc_v4l_drv
modprobe hobot_ynr_v4l2
modprobe vid_v4l2 scene=0
nohup isp_service &

```

2. 建立软链接

代码块

```
ln -s /dev/video2 /dev/video-rs-depth-0
ln -s /dev/video3 /dev/video-rs-color-0
ln -s /dev/video4 /dev/video-rs-depth-1
ln -s /dev/video5 /dev/video-rs-color-1

ln -s /dev/video0 /dev/video-rs-imu-0
ln -s /dev/video0 /dev/video-rs-imu-1

```

3. 枚举设备

代码块

```
source /opt/tros/humble/setup.bash
rs-enumerate-devices
```

```
^Croot@ubuntu:~# rs-enumerate-devices
Device info:
    Name                          :     Intel RealSense D457
    Serial Number                 :     239222300221
    Firmware Version              :     5.17.0.10
    Recommended Firmware Version  :     5.17.0.10
    Physical Port                 :     /dev/video-rs-depth-0
    Debug Op Code                 :     15
    Advanced Mode                 :     YES
    Product Id                    :     ABCD
    Camera Locked                 :     YES
    Product Line                  :     D400
    Asic Serial Number            :     318343110023
    Firmware Update Id            :     318343110023
    Dfu Device Path               :
    Connection Type               :     GMSL
```

```
Device info:
    Name                          :     Intel RealSense D457
    Serial Number                 :     243122301245
    Firmware Version              :     5.17.0.10
    Recommended Firmware Version  :     5.17.0.10
    Physical Port                 :     /dev/video-rs-depth-1
    Debug Op Code                 :     15
    Advanced Mode                 :     YES
    Product Id                    :     ABCD
    Camera Locked                 :     YES
    Product Line                  :     D400
    Asic Serial Number            :     304143110047
    Firmware Update Id            :     304143110047
    Dfu Device Path               :
    Connection Type               :     GMSL
```

4. ROS2驱动启动相机

代码块

```
source /opt/ros/humble/setup.bash
ros2 launch realsense2_camera rs_launch.py serial_no:="\"239222300221\"" camera_name:=cam1 camera_namespace:=realsense enable_color:=true enable_depth:=true align_depth.enable:=true rgb_camera.color_profile:=1280,720,15 depth_module.depth_profile:=1280,720,15

source /opt/ros/humble/setup.bash
ros2 launch realsense2_camera rs_launch.py serial_no:="\"243122301245\"" camera_name:=cam2 camera_namespace:=realsense enable_color:=true enable_depth:=true align_depth.enable:=true rgb_camera.color_profile:=1280,720,15 depth_module.depth_profile:=1280,720,15
```

```
root@ubuntu:~# ros2 topic hz /realsense/cam1/depth/image_rect_raw
average rate: 14.996
        min: 0.045s max: 0.089s std dev: 0.00790s window: 16
average rate: 14.993
        min: 0.045s max: 0.089s std dev: 0.00568s window: 31
average rate: 14.990
        min: 0.041s max: 0.093s std dev: 0.00720s window: 46
average rate: 14.991
        min: 0.041s max: 0.093s std dev: 0.00625s window: 61
average rate: 14.991
        min: 0.038s max: 0.095s std dev: 0.00726s window: 76
average rate: 14.991
        min: 0.038s max: 0.095s std dev: 0.00664s window: 91
```

```
^Croot@ubuntu:~# ros2 topic hz /realsense/cam2/depth/image_rect_raw
average rate: 14.999
        min: 0.065s max: 0.067s std dev: 0.00038s window: 16
average rate: 14.995
        min: 0.046s max: 0.087s std dev: 0.00527s window: 31
average rate: 14.992
        min: 0.046s max: 0.087s std dev: 0.00474s window: 46
average rate: 14.991
        min: 0.042s max: 0.091s std dev: 0.00603s window: 61
average rate: 14.991
        min: 0.042s max: 0.091s std dev: 0.00540s window: 76
average rate: 14.990
        min: 0.042s max: 0.091s std dev: 0.00494s window: 91
average rate: 14.990
        min: 0.042s max: 0.091s std dev: 0.00462s window: 106
```

```
^Croot@ubuntu:~# ros2 topic hz /realsense/cam1/color/image_raw
average rate: 14.999
        min: 0.043s max: 0.091s std dev: 0.00849s window: 16
average rate: 14.993
        min: 0.043s max: 0.091s std dev: 0.00610s window: 31
average rate: 14.992
        min: 0.041s max: 0.093s std dev: 0.00743s window: 46
average rate: 14.991
        min: 0.041s max: 0.093s std dev: 0.00645s window: 61
average rate: 14.989
        min: 0.037s max: 0.096s std dev: 0.00750s window: 76
```

```
^Croot@ubuntu:~# ros2 topic hz /realsense/cam2/color/image_raw
average rate: 14.999
        min: 0.065s max: 0.068s std dev: 0.00053s window: 16
average rate: 14.994
        min: 0.065s max: 0.068s std dev: 0.00038s window: 31
average rate: 14.992
        min: 0.038s max: 0.092s std dev: 0.00575s window: 46
average rate: 14.991
        min: 0.038s max: 0.092s std dev: 0.00517s window: 61
average rate: 14.990
        min: 0.038s max: 0.092s std dev: 0.00478s window: 76
average rate: 14.990
        min: 0.038s max: 0.092s std dev: 0.00437s window: 91
average rate: 14.990
        min: 0.038s max: 0.092s std dev: 0.00405s window: 106
```

三、D457相机*4启动

当前s100中无其他相机设备，若连接了其他相机设备需修改对应设备号

1. 切换到v4l2模式

代码块

```
#卸载hbn驱动
rmmod hobot_isp
rmmod hobot_cim
rmmod hobot_mipidbg
rmmod hobot_mipicsi
rmmod hobot_pym_jplus
rmmod hobot_gdc
rmmod hobot_ynr

#加载v4l2驱动
modprobe videobuf2-common

modprobe videobuf2-v4l2

modprobe videobuf2-memops

modprobe videobuf2-dma-contig

modprobe v4l2-mem2mem

modprobe max9295a
modprobe max96712
modprobe d457

modprobe v4l_mipicsi

modprobe v4l2_cim

modprobe hobot_isp_v4l2

modprobe pym_v4l_drv
modprobe gdc_v4l_drv
```

```
modprobe hobot_ynr_v4l2
modprobe vid_v4l2 scene=21
nohup isp_service &

```

2. 查看 video 节点并建立软链接

D457 硬件连接及上述 scene 选择不同，video 节点会变化，但是如果系统插入 usb camera 或者其他 video 设备，枚举出来的 video 节点可能会变化，可以通过 media-ctl -p 命令查看。

代码块

```
media-ctl -p
```

还可以通过查看 video 设备的 formats 来确定 D457 是否枚举成功。

代码块

```
v4l2-ctl -d 4 --list-formats
```

因为 librealsense SDK 只识别指定的 video 节点名称，所以需要建立软链接，以 4v rgb + depth，硬件连接为 RX4 PortAC + RX1 PortAC 为例。

video 节点	软链接(4v depth + rgb)
video4	video-rs-depth-0
video5	video-rs-color-0
video6	video-rs-depth-1
video7	video-rs-color-1
video8	video-rs-depth-2
video9	video-rs-color-2
video10	video-rs-depth-3
video11	video-rs-color-3

代码块

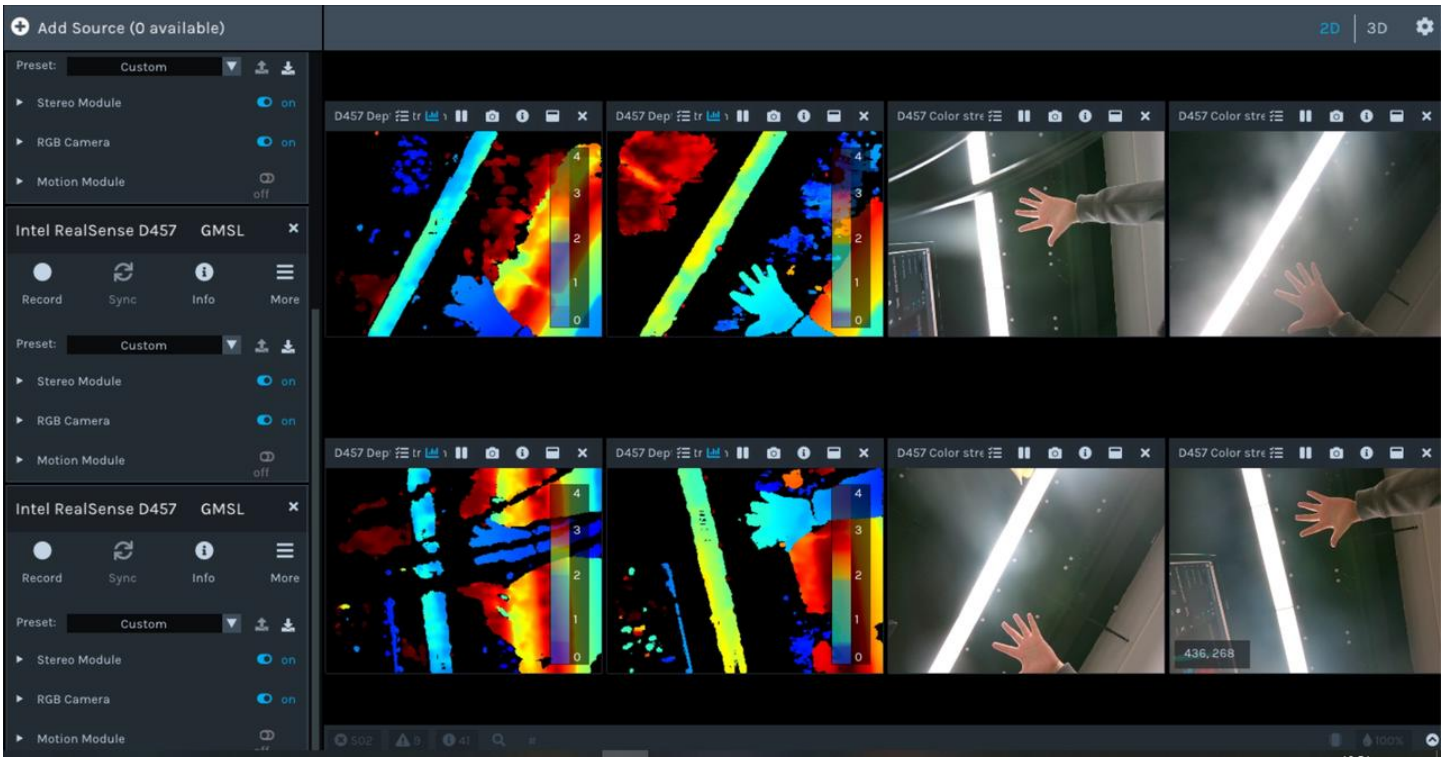
```
ln -s /dev/video4 /dev/video-rs-depth-0
```

```
ln -s /dev/video5 /dev/video-rs-color-0
ln -s /dev/video6 /dev/video-rs-depth-1
ln -s /dev/video7 /dev/video-rs-color-1
ln -s /dev/video8 /dev/video-rs-depth-2
ln -s /dev/video9 /dev/video-rs-color-2
ln -s /dev/video10 /dev/video-rs-depth-3
ln -s /dev/video11 /dev/video-rs-color-3
# video0 为 RX0，目前没有使用。
# 增加 imu 软链接是为了 S100 平台适配 librealsense，目前无 imu 功能。
ln -s /dev/video0 /dev/video-rs-imu-0
ln -s /dev/video0 /dev/video-rs-imu-1
ln -s /dev/video1 /dev/video-rs-imu-2
ln -s /dev/video1 /dev/video-rs-imu-3
```

3. 查看设备

代码块

```
source /opt/tros/humble/setup.bash
realsense-viewer
```

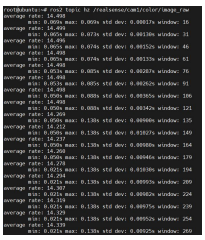
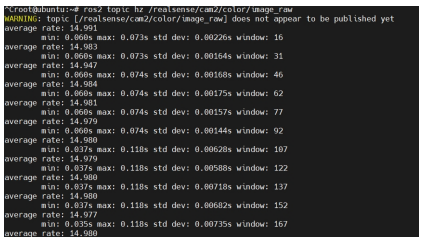
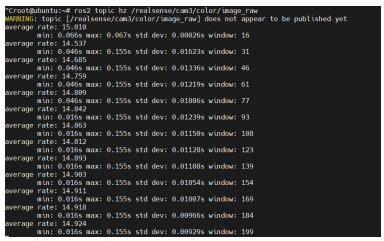
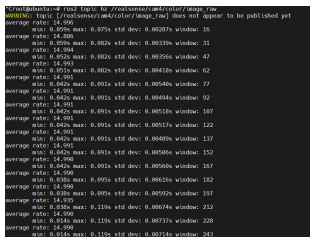
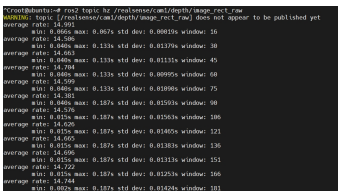
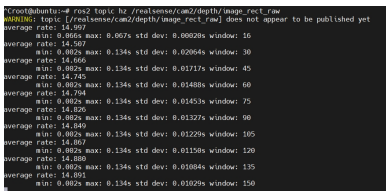
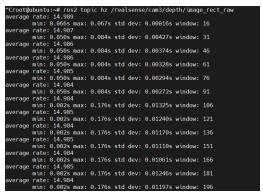
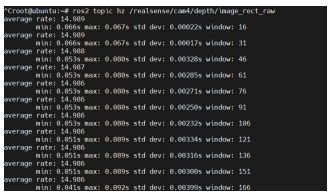


4. ROS2驱动启动相机

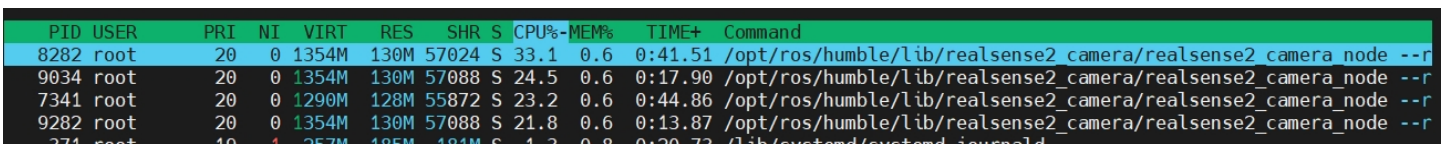
代码块

```
source /opt/ros/humble/setup.bash
ros2 launch realsense2_camera rs_launch.py serial_no:="\"239222300221\""
camera_name:=camF camera_namespace:=realsense enable_color:=true
```

```
enable_depth:=true align_depth.enable:=true
rgb_camera.color_profile:=1280,720,15 depth_module.depth_profile:=1280,720,15

source /opt/ros/humble/setup.bash
ros2 launch realsense2_camera rs_launch.py serial_no:="\"243122301245\"" camera_name:=cam2 camera_namespace:=realsense enable_color:=true enable_depth:=true align_depth.enable:=true rgb_camera.color_profile:=1280,720,15 depth_module.depth_profile:=1280,720,15

source /opt/ros/humble/setup.bash
ros2 launch realsense2_camera rs_launch.py serial_no:="\"318122301225\"" camera_name:=cam3 camera_namespace:=realsense enable_color:=true enable_depth:=true align_depth.enable:=true rgb_camera.color_profile:=1280,720,15 depth_module.depth_profile:=1280,720,15

source /opt/ros/humble/setup.bash
ros2 launch realsense2_camera rs_launch.py serial_no:="\"318122302243\"" camera_name:=cam4 camera_namespace:=realsense enable_color:=true enable_depth:=true align_depth.enable:=true rgb_camera.color_profile:=1280,720,15 depth_module.depth_profile:=1280,720,15
```

输出频率如上图所示，可满足需求，此时占用情况如下图所示，四相机同时启动占用约为1个全核CPU



PID	USER	PRI	NI	VIRT	RES	SHR	S	CPU%	MEM%	TIME+	Command
8282	root	20	0	1354M	130M	57024	S	33.1	0.6	0:41.51	/opt/ros/humble/lib/realsense2_camera/realsense2_camera_node --r
9034	root	20	0	1354M	130M	57088	S	24.5	0.6	0:17.90	/opt/ros/humble/lib/realsense2_camera/realsense2_camera_node --r
7341	root	20	0	1290M	128M	55872	S	23.2	0.6	0:44.86	/opt/ros/humble/lib/realsense2_camera/realsense2_camera_node --r
9282	root	20	0	1354M	130M	57088	S	21.8	0.6	0:13.87	/opt/ros/humble/lib/realsense2_camera/realsense2_camera_node --r

三号机器

Serial Number : 338122301538

Serial Number : 338122302167 后视

Serial Number : 338122301605 左

Serial Number : 338122301245 前视

代码块

```
source /opt/ros/humble/setup.bash
ros2 launch realsense2_camera rs_launch.py serial_no:="\"338122301245\"" camera_name:=camF camera_namespace:=realsense enable_color:=true enable_depth:=true align_depth.enable:=true rgb_camera.color_profile:=1280,720,15 depth_module.depth_profile:=1280,720,15

source /opt/ros/humble/setup.bash
ros2 launch realsense2_camera rs_launch.py serial_no:="\"338122302167\"" camera_name:=camB camera_namespace:=realsense enable_color:=true enable_depth:=true align_depth.enable:=true rgb_camera.color_profile:=1280,720,15 depth_module.depth_profile:=1280,720,15

source /opt/ros/humble/setup.bash
ros2 launch realsense2_camera rs_launch.py serial_no:="\"318122301225\"" camera_name:=cam3 camera_namespace:=realsense enable_color:=true enable_depth:=true align_depth.enable:=true rgb_camera.color_profile:=1280,720,15 depth_module.depth_profile:=1280,720,15

source /opt/ros/humble/setup.bash
ros2 launch realsense2_camera rs_launch.py serial_no:="\"318122302243\"" camera_name:=cam4 camera_namespace:=realsense enable_color:=true enable_depth:=true align_depth.enable:=true rgb_camera.color_profile:=1280,720,15 depth_module.depth_profile:=1280,720,15
```

一号机器

Serial Number : 338122303158

Serial Number : 338122301240

Serial Number : 338122303185

Serial Number : 338122302184

代码块

```
source /opt/ros/humble/setup.bash
ros2 launch realsense2_camera rs_launch.py serial_no:="\"338122303158\"" camera_name:=cam1 camera_namespace:=realsense enable_color:=true enable_depth:=true align_depth.enable:=true rgb_camera.color_profile:=1280,720,15 depth_module.depth_profile:=1280,720,15

source /opt/ros/humble/setup.bash
```

```
ros2 launch realsense2_camera rs_launch.py serial_no:="\"338122301240\"" camera_name:=cam2 camera_namespace:=realsense enable_color:=true enable_depth:=true align_depth.enable:=true rgb_camera.color_profile:=1280,720,15 depth_module.depth_profile:=1280,720,15

source /opt/ros/humble/setup.bash
ros2 launch realsense2_camera rs_launch.py serial_no:="\"338122303185\"" camera_name:=cam3 camera_namespace:=realsense enable_color:=true enable_depth:=true align_depth.enable:=true rgb_camera.color_profile:=1280,720,15 depth_module.depth_profile:=1280,720,15

source /opt/ros/humble/setup.bash
ros2 launch realsense2_camera rs_launch.py serial_no:="\"338122302184\"" camera_name:=cam4 camera_namespace:=realsense enable_color:=true enable_depth:=true align_depth.enable:=true rgb_camera.color_profile:=1280,720,15 depth_module.depth_profile:=1280,720,15
```

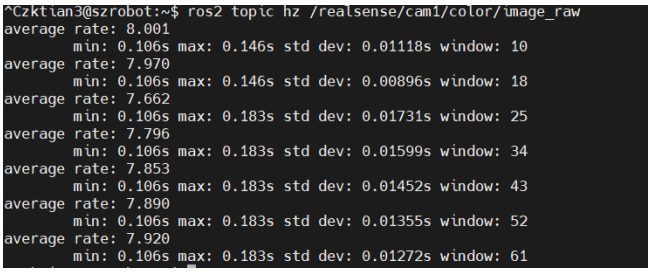
Serial Number	:	338122303633
Serial Number	:	338122300622
Serial Number	:	338122300575
Serial Number	:	338122302187

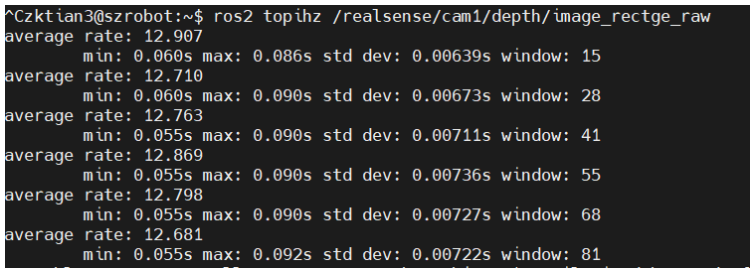
测试3号机RGBD数据（WIFI）--->144服务器（有线连接路由器），出现异常如下：

1. 存在压缩数据，但压缩数据内容为空；

```
/realsense/cam333/aligned_depth_to_color/camera_info
/realsense/cam333/aligned_depth_to_color/image_raw
/realsense/cam333/aligned_depth_to_color/image_raw/compressed
/realsense/cam333/aligned_depth_to_color/image_raw/compressedDepth
/realsense/cam333/aligned_depth_to_color/image_raw/theora
/realsense/cam333/color/camera_info
/realsense/cam333/color/image_raw
/realsense/cam333/color/image_raw/compressed
/realsense/cam333/color/image_raw/compressedDepth
/realsense/cam333/color/image_raw/theora
/realsense/cam333/color/metadata
/realsense/cam333/depth/camera_info
/realsense/cam333/depth/image_rect_raw
/realsense/cam333/depth/image_rect_raw/compressed
/realsense/cam333/depth/image_rect_raw/compressedDepth
/realsense/cam333/depth/image_rect_raw/theora
/realsense/cam333/depth/metadata
/realsense/cam333/extrinsics/depth_to_color
/realsense/cam333/extrinsics/depth_to_depth
```

2. 在服务器中订阅深度及RGB数据时，**延迟极大，订阅帧率约为0.3-0.5hz**，同步使用另一s100连接WiFi启动相机驱动，此时在144服务器中订阅帧率如下，RGB帧率为8hz左右，深度数据频率为12hz左右





四、相机脚本启动部署

camera_start.service
246 B

代码块

```
[Unit]
Description=Camera Drivers Initialization Service
After=network.target

[Service]
Type=forking
ExecStart=/usr/local/bin/camera_start.sh
RemainAfterExit=yes
StandardOutput=journal
StandardError=journal

[Install]
WantedBy=multi-user.target
```

启动脚本中对应的上环视相机设备号需结合真机进行修改，当前设备号为仅安装上环视相机情况；

camera_start.sh

代码块

```
#部署脚本
sudo cp camera_start.sh /usr/local/bin/
sudo chmod 777 /usr/local/bin/camera_start.sh
#部署服务
sudo cp camera_start.service /etc/systemd/system/
sudo chmod 777 /etc/systemd/system/camera_start.service
#应用开机自启
sudo systemctl daemon-reload
sudo systemctl enable camera_start.service
#测试服务
sudo systemctl start camera_start.service
sudo systemctl status camera_start.service
```

代码块

```
source /home/jwchen21/.bashrc
conda activate /mnt2/jwchen21/conda/asm-lab/
```

五、机器序列号

四号真机

```
338122300622左
338122300575后
338122303633右
338122302187前
```

五号真机

```
338122300982前
338122301640右
338122300545后
338122301405左
```

六号真机

```
338122300743前
338122303269右
338122300617后
338122300890左
```

2026-02-06 10:50:13.612 - MainThread - NavAgent - INFO - === Target hat Processed ===

2026-02-06 10:50:13.612 - MainThread - NavAgent - INFO - Camera: camera1

2026-02-06 10:50:13.612 - MainThread - NavAgent - INFO - BBox: x=611.0, y=470.0, w=146.0, h=110.0

2026-02-06 10:50:13.612 - MainThread - NavAgent - INFO - Center Pixel: (684, 525)

2026-02-06 10:50:13.612 - MainThread - NavAgent - INFO - Depth: 0.87m

2026-02-06 10:50:13.612 - MainThread - NavAgent - INFO - Camera: (0.05, 0.21, 0.87)

2026-02-06 10:50:13.612 - MainThread - NavAgent - INFO - Robot: pos=(-1.83, -3.36), yaw=-178.9°

2026-02-06 10:50:13.612 - MainThread - NavAgent - INFO - World: (-2.69, -3.33)

2026-02-06 10:50:13.612 - MainThread - NavAgent - INFO - Map: (566, 553)

2026-02-06 10:50:13.612 - MainThread - NavAgent - INFO - Target Yaw: -178.9°

2026-02-06 10:51:19.157 - MainThread - NavAgent - INFO - === Target hat Processed ===

2026-02-06 10:51:19.158 - MainThread - NavAgent - INFO - Camera: camera4

2026-02-06 10:51:19.158 - MainThread - NavAgent - INFO - BBox: x=453.0, y=473.0, w=148.0, h=111.0

2026-02-06 10:51:19.158 - MainThread - NavAgent - INFO - Center Pixel: (527, 528)

2026-02-06 10:51:19.158 - MainThread - NavAgent - INFO - Depth: 0.85m

2026-02-06 10:51:19.158 - MainThread - NavAgent - INFO - Camera: (-0.16, 0.21, 0.85)

2026-02-06 10:51:19.158 - MainThread - NavAgent - INFO - Robot: pos=(-1.90, -3.29), yaw=-98.3°

2026-02-06 10:51:19.158 - MainThread - NavAgent - INFO - World: (-1.04, -3.25)

2026-02-06 10:51:19.158 - MainThread - NavAgent - INFO - Map: (565, 520)

2026-02-06 10:51:19.158 - MainThread - NavAgent - INFO - Target Yaw: 171.7°

2026-02-06 10:51:19.158 - MainThread - NavAgent - WARNING - Star position out of map image: (565, 520)

2026-02-06 10:52:11.526 - MainThread - NavAgent - INFO - === Target hat Processed ===

2026-02-06 10:52:11.526 - MainThread - NavAgent - INFO - Camera: camera2

2026-02-06 10:52:11.526 - MainThread - NavAgent - INFO - BBox: x=452.0, y=474.0, w=148.0, h=112.0

2026-02-06 10:52:11.526 - MainThread - NavAgent - INFO - Center Pixel: (526, 530)

2026-02-06 10:52:11.526 - MainThread - NavAgent - INFO - Depth: 0.84m

2026-02-06 10:52:11.526 - MainThread - NavAgent - INFO - Camera: (-0.15, 0.21, 0.84)

2026-02-06 10:52:11.526 - MainThread - NavAgent - INFO - Robot: pos=(-1.79, -3.31), yaw=-9.1°

2026-02-06 10:52:11.526 - MainThread - NavAgent - INFO - World: (-2.65, -3.33)

2026-02-06 10:52:11.526 - MainThread - NavAgent - INFO - Map: (566, 552)

2026-02-06 10:52:11.526 - MainThread - NavAgent - INFO - Target Yaw: 170.9°

2026-02-06 10:52:11.526 - MainThread - NavAgent - WARNING - Star position out of map image: (566, 552)

2026-02-06 10:52:11.526 - MainThread - NavAgent - INFO - === Target hat Processed ===

2026-02-06 10:52:11.526 - MainThread - NavAgent - INFO - Camera: camera2

2026-02-06 10:52:11.526 - MainThread - NavAgent - INFO - BBox: x=452.0, y=474.0, w=148.0, h=112.0

2026-02-06 10:52:11.526 - MainThread - NavAgent - INFO - Center Pixel: (526, 530)

2026-02-06 10:52:11.526 - MainThread - NavAgent - INFO - Depth: 0.84m

2026-02-06 10:52:11.526 - MainThread - NavAgent - INFO - Camera: (-0.15, 0.21, 0.84)

2026-02-06 10:52:11.526 - MainThread - NavAgent - INFO - Robot: pos=(-1.79, -3.31), yaw=-9.1°

2026-02-06 10:52:11.526 - MainThread - NavAgent - INFO - World: (-2.65, -3.33)

2026-02-06 10:52:11.526 - MainThread - NavAgent - INFO - Map: (566, 552)

2026-02-06 10:52:11.526 - MainThread - NavAgent - INFO - Target Yaw: 170.9°

2026-02-06 10:52:11.526 - MainThread - NavAgent - WARNING - Star position out of map image: (566, 552)